HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Engineered **Python** scripts to automate data extraction and transformation for the financial-services operations dashboard, reducing manual processing time by approximately **15%**.
- Developed and maintained **Java Spring Boot** REST APIs for cross-system integrations, supporting internal business applications with a focus on clear error handling and efficient data exchange.
- Managed production incidents and performed root cause analysis for enterprise applications using **Splunk** and **Dynatrace**, adhering to strict SLAs for P1 and P2 issues.
- Contributed to the development of a **GPU kernel** for Blackwell architecture, focusing on optimizing FP8 paths to improve inference throughput by **8%** on specific model types.
- Debugged and resolved complex issues spanning **GPU memory**, the engine scheduler, and container configurations, documenting findings for team review.
- Implemented optimizations for **transformer inference**, specifically tuning KV cache behavior and continuous batching for new models, leading to a measurable increase in goodput per GPU.
- Containerized inference engines using **Docker** and deployed them to a **Kubernetes** cluster, managing configurations and ensuring consistent rollouts across the fleet.

SKILLS

Programming – Python, Java, C++, CUDA

AI/ML – Transformer Inference, GPU Kernels, FP8, KV Cache, Continuous Batching, Model Bring-up, Quantisation

Cloud/Containerization – Docker, Kubernetes, AWS (EC2, S3, Lambda, CloudWatch)

Databases – PostgreSQL, MySQL, Oracle, SQL

Tools – Git, Jenkins, Splunk, Dynatrace, Jira, Confluence, PagerDuty

Frameworks – Spring Boot, FastAPI, SGLang, vLLM, TensorRT-LLM

Methodologies – Agile, Scrum, CI/CD

PROJECTS

GPU Kernel Optimization for LLM Inference

- Developed and benchmarked a custom **CUDA kernel** to accelerate FP8 computations within a transformer model's attention mechanism, achieving a **12%** speedup on a Blackwell GPU.
- Integrated the optimized kernel into the **SGLang** inference engine, performing rigorous testing against production traffic patterns to validate performance gains and stability.
- Documented the kernel's implementation, profiling results, and trade-offs, enabling the Inference Team to confidently adopt the change for new model deployments.

Automated Data Pipeline for Financial Operations

- Built a **Python**-based ETL pipeline to ingest and process transaction data from multiple sources for a financial-services operations dashboard, improving data availability for monitoring.
- Implemented error handling and logging within the pipeline using **Pandas** and **SQLAlchemy**, ensuring data integrity and providing clear audit trails for critical financial operations.
- Scheduled and monitored pipeline execution via **cron jobs** on Linux, ensuring timely delivery of processed data for operational reporting and alerting.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)